

THIRD SEMESTER**B.E. (IT)**

B.E. END SEM. EXAMINATION, Nov.-2018

ITC07: Digital Circuits and Systems

Time: 03:00 Hrs.

Max. Marks: 40

Note: Attempt **ANY EIGHT** questions. All questions carry equal marks. Assume ~~any~~ missing data, if any.

1. Identify the prime and essential prime implicants for the following function and simplify it using Quine-McClusky (Tabulation method). [5]
 $S = f(A, B, C, D) = \pi M(5, 7, 8, 21, 23, 26, 30) + \pi D(10, 14, 24, 28)$
2. Design a combinational circuit to find the 9's complement of a BCD number and implement the circuit using 8X1 Mux. [5]
3. (a) An 8X1 mux has inputs A, B, and C connected to the selection inputs S2, S1 and S0 respectively. The data inputs I0 to I7 are as follows: [3]
 - i. $I0 = I4 = \overline{D}$; $I1 = I2 = I6 = 0$; $I3 = I5 = 1$ and $I7 = D$
 - ii. $I0 = I6 = 1$; $I3 = I7 = 1$; $I1 = I2 = \overline{D}$; $I4 = I5 = D$

Determine the Boolean function that the multiplexer implements.

(b) Implement 8X1 Mux with two 4X1 multiplexers. Use external gates if necessary. [2]
4. (a) Draw the circuit of a positive edge triggered J-K flip-flop and explain its operation with the help of truth table. How is the race around condition eliminated? [3]
 (b) Convert J-K flip-flop to S-R flip-flop. [2]
5. Using a positive edge triggered AB flip-flop (characteristic table given below) design a synchronous mod-5 counter whose counting sequence is: [5]

 $110 \rightarrow 101 \rightarrow 001 \rightarrow 111 \rightarrow 011 \rightarrow 110$

Design the circuit so that whenever it goes to an invalid state it comes back to state 111.

Table -1 characteristic table of AB flip-flop

A	B	Q(t+1)
0	0	1
0	1	Q(t)
1	0	$\overline{Q(t)}$
1	1	0

6. Design a register, incorporating 4 multiplexers and 4 positive edge triggered D flip-flop, having the behavior specified by table 2. [5]

Table -2 Table for shift register

Select Line S0	Select Line S1	Register Operation
0	0	Circular shift right
0	1	Hold
1	0	Synchronous Reset
1	1	Compliment Content

7. (a) With the help of suitable neat diagram explain the functioning of a TTL gate with totem pole output configuration. How is propagation delay reduced by this configuration? [3]
 (b) Explain the following parameters: [2]
 - i. Propagation Delay
 - ii. Noise Margin

- iii. Fan-out
- iv. Power dissipation

8. Explain with the help of diagram the dual slope A/D converter. Also convert 6.2 V into 4-bit binary data and mention the time taken to convert it, if time period of a clock pulse is 5ns. [3+2]
9. Draw an ASM chart, state diagram and state table for the circuit shown in Fig. 1. Also show its implementation using T-FF and Mux. [3+2]

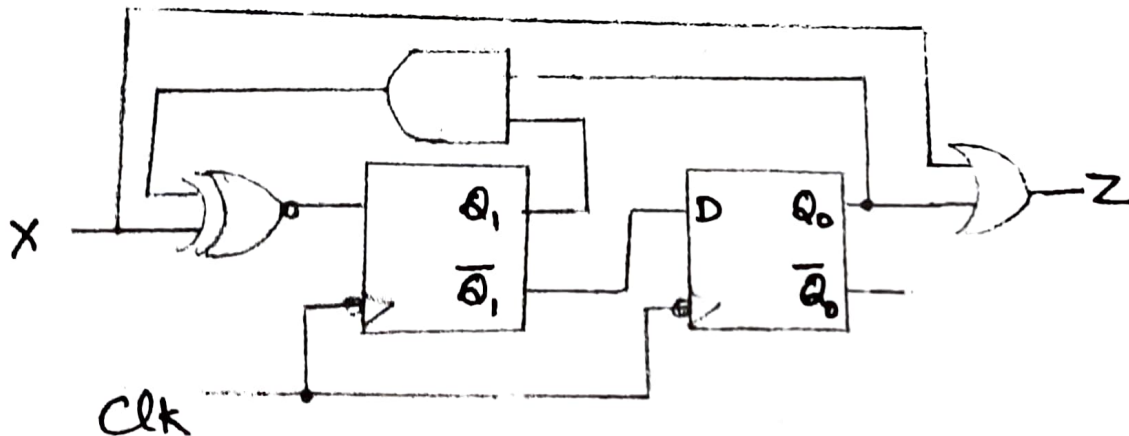


Fig 1

10. (a) Describe the principle of Astable multivibrator implemented using gates. [3]
- (b) Explain the basic static RAM cell and dynamic RAM cell. [2]